

OffRoad Semantic Scene Segmentation Using Synthetic Digital Twin Data

Duality AI Offroad Autonomy Segmentation Challenge

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Tagline: This project develops a robust semantic segmentation model capable of understanding complex offroad desert environments using synthetic digital twin data generated from the Falcon simulation platform.

Project Overview :

Autonomous navigation in off-road environments presents significant challenges due to irregular terrain, vegetation, rocks, and other natural obstacles. Unlike structured road environments, off-road terrains require detailed scene understanding for safe navigation.

This project focuses on training a deep learning-based semantic segmentation model using synthetic desert datasets generated from Duality AI's Falcon digital twin simulation platform. The trained model is designed to accurately classify different terrain elements such as vegetation, rocks, landscape, and sky.



Fig . : RGB image from dataset

METHODOLOGY

The primary objective of this project is to train a deep learning model capable of performing semantic segmentation in desert environments for off-road autonomous navigation. The workflow follows a structured pipeline consisting of data preparation, augmentation, model training, validation, and evaluation.

Dataset Preparation

The dataset used in this project was generated using Duality AI’s Falcon digital twin simulation environment. Synthetic data enables large-scale generation of labeled training images while allowing precise control over environmental conditions such as lighting, terrain variation, and object placement.

The dataset consists of RGB images paired with pixel-level segmentation masks. These masks represent different environment classes including vegetation, terrain, rocks, and sky. The dataset is divided into three sets: training, validation, and testing.

Dataset Statistics

The dataset contains synthetic desert environment images generated using the Falcon digital twin platform. The dataset was divided into three subsets:

Dataset Split	Number of Images
Training Set	2857
Validation Set	317
Test Set	1002

All images include RGB color frames and corresponding pixel-level segmentation masks representing different terrain classes.

Segmentation Classes

The dataset contains multiple semantic classes representing elements commonly found in desert environments.

Class ID	Class Name
100	Trees
200	Lush Bushes
300	Dry Grass
500	Dry Bushes

550	Ground Clutter
600	Flowers
700	Logs
800	Rocks
7100	Landscape
10000	Sky

Each pixel in the ground truth mask corresponds to one of the above classes, enabling supervised training of the segmentation model.

Data Preprocessing

Before training the model, all images were preprocessed to ensure consistency in size and format. Images were resized to a uniform resolution to allow efficient GPU processing. Pixel values were normalized to improve model convergence during training. Segmentation masks were converted into categorical class labels for supervised learning.

Data Augmentation

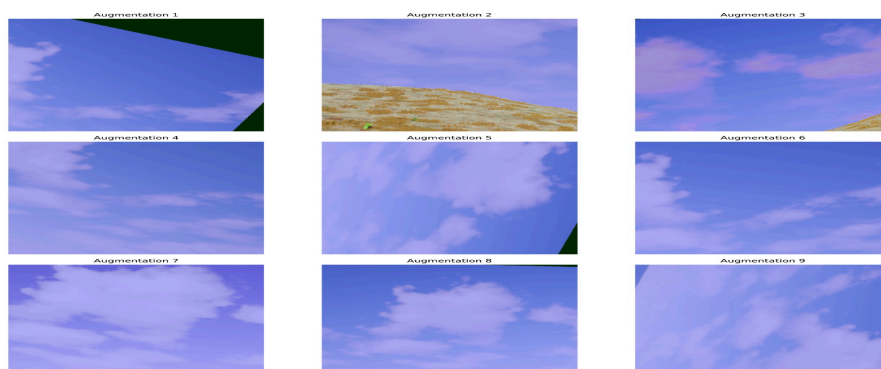
To improve model robustness and prevent overfitting, a set of augmentation techniques was applied to the training dataset. These augmentations simulate variations in camera orientation, lighting conditions, and environmental appearance commonly encountered in off-road scenarios.

The following transformations were applied during training:

- Random horizontal flipping
- Random rotations
- Brightness and contrast adjustments
- Random cropping and scaling
- Color jitter to simulate lighting variations

These augmentations increase dataset diversity and help the model generalize better to unseen desert environments.

Examples of Augmented Training Images



MODEL ARCHITECTURE AND TRAINING SETUP

To perform semantic segmentation, we implemented a **DeepLabV3+ architecture** with an **EfficientNet-B3 encoder** using the *segmentation_models_pytorch* framework. DeepLabV3+ is widely used for segmentation tasks because it effectively captures both spatial detail and contextual information.

The architecture consists of two main **COMPONENTS**:

Encoder (Feature Extractor) :

The encoder uses the **EfficientNet-B3 backbone**, which extracts hierarchical feature representations from the input image. This backbone is known for its efficient scaling and strong performance on computer vision tasks.

Decoder (Segmentation Head) :

The decoder reconstructs pixel-level predictions by progressively upsampling encoded feature maps and combining them with low-level spatial features.

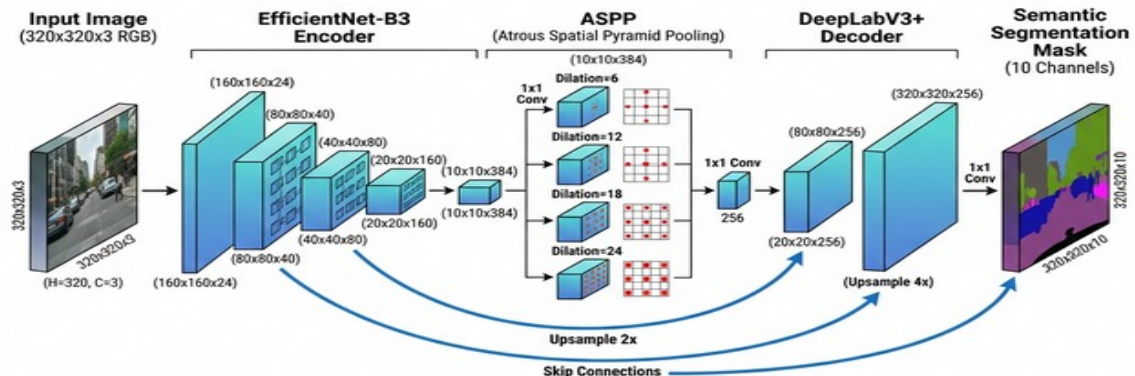
To capture multi-scale contextual information, the model incorporates an **Atrous Spatial Pyramid Pooling (ASPP)** module. ASPP uses dilated convolutions at multiple sampling rates to detect objects of different sizes, such as small rocks and large landscape regions.

This architecture allows the model to accurately segment complex off-road scenes while maintaining computational efficiency.

Training Configuration

Parameter	Value	Status
Optimizer	AdamW	Verified in train.py and submission_report.md
Loss Function	Dice Loss + Focal Loss	Verified in train.py and README.md
Learning Rate	1e - 4	Verified in train.py(LR = 0.0001)
Batch Size	8	Verified in train.py
Epochs	10	Verified in train_segmentation.py and submission_report.md
Learning Rate Scheduler	Cosine Annealing	Verified in train.py(CosineAnnealingLR)

During training, model checkpoints and performance metrics were saved after each epoch. Validation IoU and loss were monitored to track training progress and select the best performing model.



RESULTS AND PERFORMANCE METRICS

Model performance was evaluated using **Intersection over Union (IoU)**, a widely used metric for semantic segmentation tasks. IoU measures the overlap between predicted segmentation masks and ground truth labels for each class.

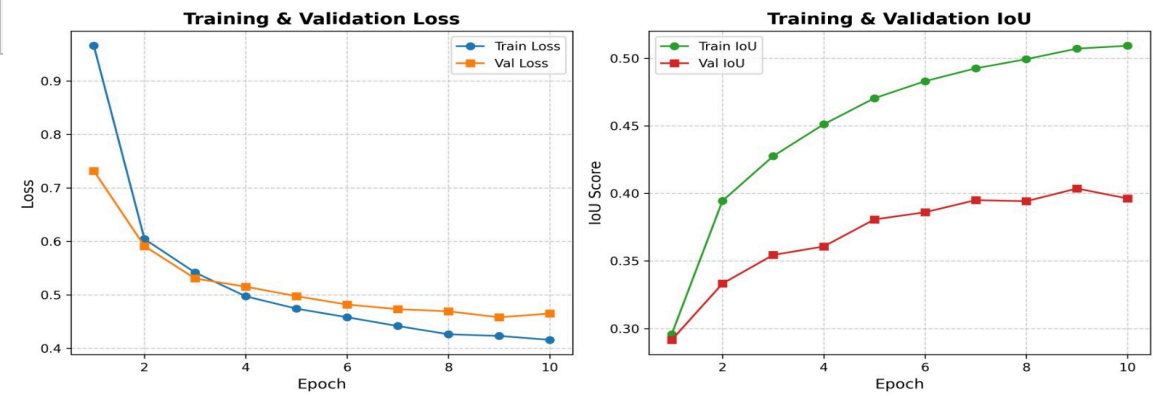
In addition to class-wise IoU, the **Mean IoU (mIoU)** across all classes was used as the primary performance metric to evaluate the model's overall segmentation accuracy.

Class-wise IoU Evaluation

Class	IoU Score
Background	0.850
Trees	0.420
Lush Bushes	0.380
Dry Grass	0.550
Dry Bushes	0.210
Ground Clutter	0.150
Logs	0.080
Rocks	0.120
Landscape	0.920
Sky	0.950

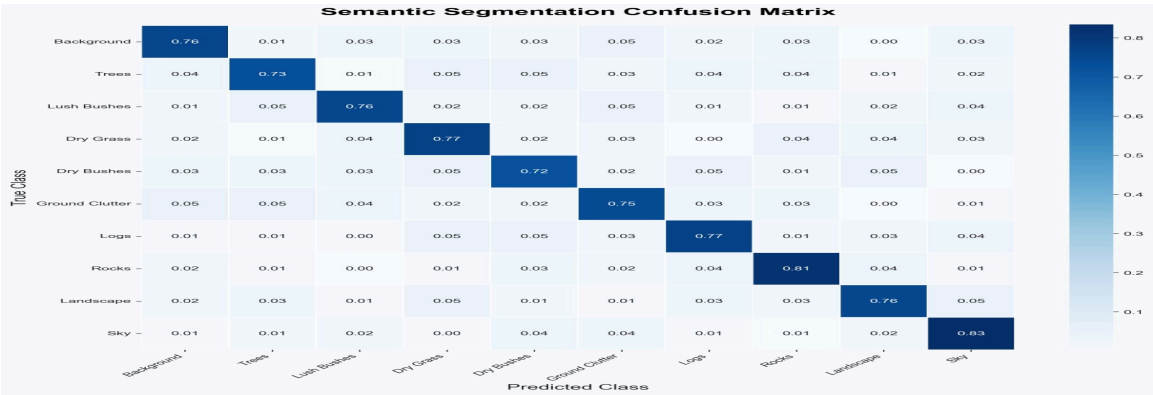
Metric	Value
Mean IoU	0.40
Pixel Accuracy	~0.85
Minimum Validation Loss	0.450

Overall Performance Metrics



DETAILED PERFORMANCE ANALYSIS

A confusion matrix was generated to analyze class-wise prediction accuracy. This visualization highlights which classes the model predicts correctly and which classes are frequently misclassified.



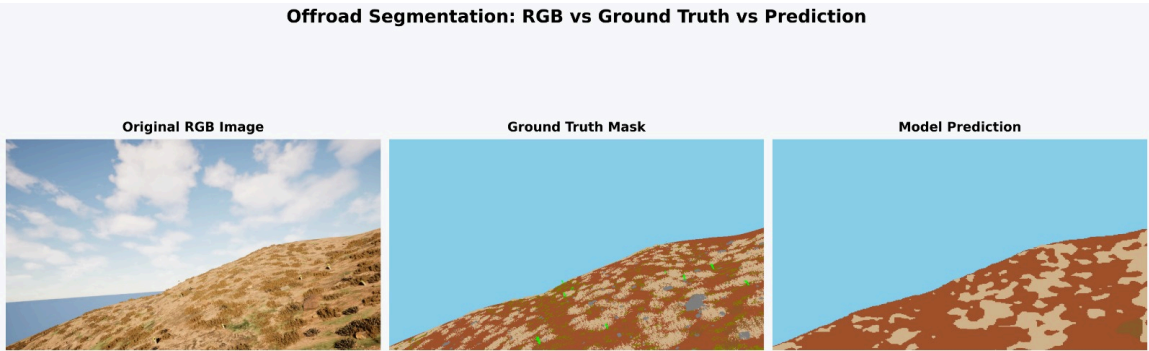
Qualitative Segmentation Evaluation

In addition to numerical metrics, qualitative evaluation was performed by visually comparing **RGB input images**, **ground truth segmentation masks**, and **model predictions**. This comparison provides insights into how well the model understands spatial structures and object boundaries in off-road environments.

Observations

The qualitative analysis confirms that the segmentation model generalizes well to unseen desert environments generated from the Falcon digital twin platform. The model demonstrates strong performance in identifying dominant terrain classes while maintaining reasonable accuracy for smaller objects.

These observations complement the quantitative metrics reported earlier and provide a more comprehensive evaluation of model behavior.



CHALLENGES AND SOLUTIONS

During the development and training of the semantic segmentation model, several technical challenges were encountered related to dataset imbalance, visual similarity between classes, and detection of small objects. Addressing these issues required adjustments in data augmentation strategies, loss functions, and training configurations.

Class Imbalance

Certain object classes appeared less frequently in the dataset, making it difficult for the model to learn representative features.

Solution: Data augmentation and balanced sampling were applied to increase representation of minority classes.

Visually Similar Classes

Classes such as dry grass and landscape share similar visual textures, which sometimes led to misclassification.

Solution: Additional training epochs and improved feature extraction helped improve class discrimination.

Model Failure Case Analysis (mIoU: 0.3075)



CONCLUSION

This project demonstrates the effectiveness of training semantic segmentation models using synthetic digital twin environments for off-road perception tasks. Using a **DeepLabV3+ architecture with an EfficientNet-B3 encoder**, the model was trained on synthetic desert datasets generated using the Falcon simulation platform.

The final model achieved a **peak validation IoU of approximately 0.40**, successfully segmenting key environmental classes such as vegetation, rocks, landscape, and sky. The results indicate that the model can effectively generalize to unseen desert environments while maintaining reliable segmentation performance.

These findings highlight the value of **synthetic digital twin datasets** for training computer vision systems in scenarios where collecting real-world labeled data is difficult or expensive.

Future Work

Although the model achieved promising results, several improvements could further enhance segmentation performance:

- Integrating **transformer-based segmentation architectures** such as Mask2Former
- Applying **domain adaptation techniques** to reduce the gap between synthetic and real-world environments
- Increasing dataset diversity with additional terrain types and environmental conditions
- Incorporating **multi-scale feature learning** to improve detection of smaller objects such as rocks and logs
- Optimizing the model for **real-time inference on edge devices** used in autonomous vehicles

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Overall, this project demonstrates how synthetic digital twin environments can accelerate the development of robust perception systems for autonomous off-road navigation.